

Teacher Information, Course Description and  
Lecture-wise Plan

of

Database Systems (CSC270)

under

Pilot Project:

Uniform Teaching, Assessment and Quality  
Assurance Across All The Campuses

by

Faculty of Information Science and Technology  
COMSATS University Islamabad

**A: Teacher Information**

|                                |                     |
|--------------------------------|---------------------|
| Course Name: Database Systems  | Credit Hours: 3+1   |
| Course Code: CSC270            | Pre-requisite:      |
| Teacher Name:                  | Teacher E-mail:     |
| Program: BS (Computer Science) | Semester: Fall 2026 |
| Department: Computer Science   | Campus:             |

**B: Course Contents**

This course introduces the fundamental concepts of database systems. Topics include: Introduction to Databases & Information Systems; Evolution of Database Systems; Components; Architecture; Functions; Relational Model; Relational Algebra; Relational Calculus; Data Modeling; Relational Data Model; Relational Algebra & Calculus; Integrity Constraints; Conceptual Models; Entity-Relationship (E-R) Model; Enhanced E-R Model; Mapping Conceptual Schema to Relational Schema; Functional Dependency & Normalization; Structured Query Language (SQL); Views; Materialized Views; Non-Relational/No SQL Databases; MongoDB as NoSQL Database; Document Model; and Transaction Management.

**Recommended Books:****Text Books:**

1. Database systems: A Practical Approach to Design, Implementation, and Management, Thomas Connolly, Carolyn Begg, Pearson, 2015.
2. MongoDB: The Definitive Guide, Shannon Bradshaw, Eoin Brazil, Kristina Chodorow, O'Reilly Media, 2019.

**Reference Books:**

3. Fundamentals of Database Systems, Elmasri, R, Navathe, Pearson, 2016.
4. Database System Concepts, Silberschatz, Korth, Sudarshan, McGraw Hill, 2019.

### C: Lecture-wise Plan

| Week # | Lecture #   | Topic (Chapter/Topics Covered)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Practice/<br>Evaluation               | Remarks                |
|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------|
| 1      | 1<br>(Date) | <b>Introduction to Databases:</b> <ul style="list-style-type: none"> <li>• Introduction to Databases</li> <li>• Information Systems</li> <li>• Database Applications</li> <li>• Evolution of DBMS</li> <li>• File-Based Systems</li> <li>• Database Approach</li> <li>• Advantages of DBMS</li> <li>• Disadvantages of DBMS</li> </ul>                                                                                                                                                                                                 | Exercises<br>as given in<br>the books | Connolly :<br>Ch1      |
|        | 2<br>(Date) | <b>Database Approach:</b> <ul style="list-style-type: none"> <li>• Data &amp; Metadata</li> <li>• Database Environment</li> <li>• Approaches &amp; Roles</li> <li>• Components</li> <li>• Functions</li> <li>• Data Dependences</li> </ul>                                                                                                                                                                                                                                                                                             | Exercises<br>as given in<br>the books | Connolly :<br>Ch1      |
| 2      | 3<br>(Date) | <b>Database Languages &amp; Functions:</b> <ul style="list-style-type: none"> <li>• Data Definition Language (DDL)</li> <li>• Data Manipulation Language (DML)</li> <li>• Data Control &amp; Security</li> <li>• DBMS Functions</li> <li>• Database Utilities</li> <li>• Database Administration Functions</li> </ul>                                                                                                                                                                                                                  | Exercises<br>as given in<br>the books | Connolly :<br>Ch2      |
|        | 4<br>(Date) | <b>Database Architectures &amp; Data Independence</b> <ul style="list-style-type: none"> <li>• DBMS Architecture</li> <li>• Two-Tier Client–Server Architecture</li> <li>• Three-Tier Client–Server Architecture</li> <li>• Three-Schema Architecture</li> <li>• External Level</li> <li>• Conceptual Level</li> <li>• Internal Level</li> <li>• External–Conceptual Mapping</li> <li>• Conceptual–Internal Mapping</li> <li>• Data Independence</li> <li>• Logical Data Independence</li> <li>• Physical Data Independence</li> </ul> | <b>Quiz # 1</b>                       | Connolly :<br>Ch2, Ch3 |
| 3      | 5<br>(Date) | <b>Relational Model:</b> <ul style="list-style-type: none"> <li>• Overview of the Relational Model</li> <li>• Notion of a Relation</li> <li>• Mathematical Relations</li> <li>• Database Relations</li> <li>• Relation Properties</li> <li>• Attributes and Domains</li> <li>• Relational Keys</li> </ul>                                                                                                                                                                                                                              | Exercises<br>as given in<br>the books | Connolly :<br>Ch4      |

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|   |              | <ul style="list-style-type: none"> <li>• Candidate, Primary &amp; Foreign Keys</li> <li>• Relational Database Schema</li> <li>• Representing Relational Database Schemas</li> </ul>                                                                                                                                                                                                                     |                                                   |                 |
|   | 6<br>(Date)  | <b>Integrity Constraints:</b> <ul style="list-style-type: none"> <li>• Domain Constraints</li> <li>• Entity Integrity</li> <li>• Referential Integrity</li> <li>• General Constraints</li> <li>• Integrity Constraints in Relational Databases</li> <li>• Enforcement of Integrity Constraints</li> </ul>                                                                                               | Exercises as given in the books                   | Connolly : Ch4  |
| 4 | 7<br>(Date)  | <b>Relational Algebra:</b> <ul style="list-style-type: none"> <li>• Introduction to Relational Algebra</li> <li>• Unary Relational Operations <ul style="list-style-type: none"> <li>○ SELECT</li> <li>○ PROJECT</li> </ul> </li> <li>• Set Operations <ul style="list-style-type: none"> <li>○ UNION</li> <li>○ INTERSECTION</li> <li>○ DIFFERENCE</li> <li>○ CARTESIAN PRODUCT</li> </ul> </li> </ul> | Exercises as given in the books + Assignment No 1 | Connolly : Ch5  |
|   | 8            | <b>Join &amp; Division Operations:</b> <ul style="list-style-type: none"> <li>• Join Operations</li> <li>• Theta Join</li> <li>• Equijoin</li> <li>• Natural Join</li> <li>• Outer Joins</li> <li>• Division Operation</li> <li>• Examples of Join &amp; Division Operations</li> </ul>                                                                                                                 | <b>Quiz # 2</b>                                   | Connolly : Ch5  |
| 5 | 9<br>(Date)  | <b>Aggregation &amp; Grouping:</b> <ul style="list-style-type: none"> <li>• Aggregate Functions</li> <li>• Grouping Operations</li> <li>• Grouping with Aggregate Functions</li> <li>• Examples of Aggregation &amp; Grouping</li> </ul>                                                                                                                                                                | Exercises as given in the books                   | Connolly : Ch5  |
|   | 10<br>(Date) | <b>Relational Calculus:</b> <ul style="list-style-type: none"> <li>• Introduction to Relational Calculus</li> <li>• Tuple Relational Calculus (TRC)</li> <li>• Tuple Variables</li> <li>• Domain Relational Calculus (DRC)</li> <li>• Domain Variables</li> <li>• Tuple vs. Domain Relational Calculus</li> </ul>                                                                                       | Exercises as given in the books                   | Connolly : Ch5  |
| 6 | 11<br>(Date) | <b>Data Modeling &amp; Database Design:</b> <ul style="list-style-type: none"> <li>• Data Modeling</li> <li>• Conceptual Data Models</li> </ul>                                                                                                                                                                                                                                                         | <b>Quiz # 3</b>                                   | Connolly : Ch10 |

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|   |              | <ul style="list-style-type: none"> <li>• Logical Data Models</li> <li>• Relational Database Design</li> <li>• Mapping Conceptual Model to Logical Model</li> <li>• Mapping Logical Model to Relational Schema</li> <li>• Relational Database Design Considerations</li> </ul>                                                                                                                                                                         |                                                   |                 |
|   | 12<br>(Date) | <b>Entity-Relationship (ER) Model:</b> <ul style="list-style-type: none"> <li>• Introduction to the ER Model</li> <li>• Entity Types</li> <li>• Relationship Types</li> <li>• Attributes</li> <li>• Strong &amp; Weak Entity Types</li> <li>• Attributes on Relationships</li> <li>• Structural Constraints</li> <li>• Cardinality Constraints</li> <li>• Participation Constraints</li> </ul>                                                        | Exercises as given in the books                   | Connolly : Ch12 |
| 7 | 13<br>(Date) | <b>ER Modeling Issues/Problems:</b> <ul style="list-style-type: none"> <li>• Choosing Entity Types</li> <li>• Choosing Relationship Types</li> <li>• Choosing Attributes</li> <li>• Identifying Keys</li> <li>• Naming Conventions</li> <li>• Handling Complex Relationships</li> <li>• Structural Constraints</li> <li>• Avoiding Redundancy</li> <li>• Modeling Generalization &amp; Specialization</li> <li>• Common ER Modeling Errors</li> </ul> | Exercises as given in the books                   | Connolly : Ch12 |
|   | 14<br>(Date) | <b>Enhanced ER (EER) Model:</b> <ul style="list-style-type: none"> <li>• Enhanced ER Model</li> <li>• Specialization</li> <li>• Generalization</li> <li>• Superclasses &amp; Subclasses</li> <li>• Inheritance</li> <li>• Constraints on Specialization/Generalization</li> <li>• Aggregation</li> <li>• Composition</li> <li>• Categorization / Union Types</li> </ul>                                                                               | Exercises as given in the books + Assignment No 2 | Connolly : Ch13 |
| 8 | 15<br>(Date) | <b>EER Modeling Case Study:</b> <ul style="list-style-type: none"> <li>• Case Study Description</li> <li>• Identification of Entity Types</li> <li>• Identification of Relationships</li> <li>• Identification of Attributes</li> <li>• Specialization &amp; Generalization</li> <li>• Superclass/Subclass Relationships</li> <li>• Inheritance</li> <li>• Aggregation &amp; Composition</li> </ul>                                                   | Exercises as given in the books                   | Connolly : Ch13 |

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|    |              | <ul style="list-style-type: none"> <li>• EER Diagram Development</li> <li>• Validation of EER Model</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                 |                                         |
|    | 16           | <b>Mapping EER to Relational Schema:</b> <ul style="list-style-type: none"> <li>• Mapping EER Model to Relations</li> <li>• Mapping Strong Entity Types</li> <li>• Mapping Weak Entity Types</li> <li>• Mapping 1:1 Relationships</li> <li>• Mapping 1:N Relationships</li> <li>• Mapping M:N Relationships</li> <li>• Mapping Multivalued Attributes</li> <li>• Mapping Specialization &amp; Generalization</li> <li>• Mapping Categories / Union Types</li> <li>• Mapping Aggregation</li> <li>• Mapping EER Constraints to Relational Schema</li> </ul> | Exercises as given in the books | Connolly : Ch17                         |
| 9  | 17<br>(Date) | Midterm Examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Midterm (25%)</b>            |                                         |
|    | 18<br>(Date) | <b>Normalization:</b> <ul style="list-style-type: none"> <li>• Purpose of Normalization</li> <li>• Normalization in Database Design</li> <li>• Data Redundancy</li> <li>• Update Anomalies</li> <li>• Functional Dependencies</li> <li>• Functional Dependency and Normalization</li> <li>• Decomposition of Relations</li> <li>• Lossless-Join Property</li> <li>• Dependency-Preservation Property</li> <li>• Normalization Process</li> </ul>                                                                                                           | Exercises as given in the books | Connolly : Ch14                         |
| 10 | 19<br>(Date) | <b>Normalization Process:</b> <ul style="list-style-type: none"> <li>• First Normal Form (1NF)</li> <li>• Second Normal Form (2NF)</li> <li>• Third Normal Form (3NF)</li> <li>• Conversion to 1NF</li> <li>• Conversion to 2NF</li> <li>• Conversion to 3NF</li> <li>• Examples of Normalization</li> <li>• Testing for Normal Forms</li> </ul>                                                                                                                                                                                                           | Exercises as given in the books | Connolly : Ch14                         |
|    | 20<br>(Date) | <b>Advanced Normalization:</b> <ul style="list-style-type: none"> <li>• Functional Dependency Inference Rules</li> <li>• Closure of Functional Dependencies</li> <li>• Attribute Closure</li> <li>• Boyce-Codd Normal Form (BCNF)</li> <li>• Testing for BCNF</li> <li>• BCNF Decomposition</li> <li>• Lossless-Join Decomposition</li> <li>• BCNF vs. 3NF</li> </ul>                                                                                                                                                                                      | Exercises as given in the books | Connolly : Ch15 & Elmasri Chapter No 15 |

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| 11 | 21<br>(Date) | <b>BCNF Schema Design by Decomposition Algorithm:</b> <ul style="list-style-type: none"> <li>• <i>BCNF Decomposition Algorithm</i></li> <li>• <i>Identifying BCNF Violations</i></li> <li>• <i>Decomposing Relations</i></li> <li>• <i>Lossless-Join Decomposition</i></li> <li>• <i>Testing Relations for BCNF</i></li> <li>• <i>Iterative BCNF Decomposition</i></li> <li>• <i>Dependency Preservation</i></li> <li>• <i>Worked Examples of BCNF</i></li> </ul>                                                        | <i>Exercises as given in the books + Assignment No 3</i> | <i>Connolly : Ch15 &amp; Elmasri Chapter No 15</i> |
|    | 22<br>(Date) | <b>Multi-Valued Dependencies and Fourth Normal Form:</b> <ul style="list-style-type: none"> <li>• <i>Multi-Valued Dependencies (MVDs)</i></li> <li>• <i>Trivial &amp; Non-Trivial MVDs</i></li> <li>• <i>Properties of MVDs</i></li> <li>• <i>Inference Rules for MVDs</i></li> <li>• <i>Fourth Normal Form (4NF)</i></li> <li>• <i>4NF Decomposition</i></li> <li>• <i>Lossless-Join Property in 4NF</i></li> <li>• <i>Examples of 4NF Decomposition</i></li> </ul>                                                     | <i>Exercises as given in the books</i>                   | <i>Connolly : Ch15</i>                             |
| 12 | 23<br>(Date) | <b>Views &amp; Materialized Views:</b> <ul style="list-style-type: none"> <li>• <i>Introduction to Views</i></li> <li>• <i>Creating Views</i></li> <li>• <i>Querying Views</i></li> <li>• <i>Updating Views</i></li> <li>• <i>Advantages &amp; Disadvantages of Views</i></li> <li>• <i>Materialized Views</i></li> <li>• <i>Creating &amp; Refreshing Materialized Views</i></li> <li>• <i>Advantages &amp; Disadvantages of Materialized Views</i></li> </ul>                                                          | <b>Quiz # 4</b>                                          | <i>Connolly : Ch7</i>                              |
|    | 24<br>(Date) | <b>Database Authorization:</b> <ul style="list-style-type: none"> <li>• <i>Database Security &amp; Authorization</i></li> <li>• <i>Database Users &amp; Privileges</i></li> <li>• <i>Access Control</i></li> <li>• <i>Granting Privileges</i></li> <li>• <i>Revoking Privileges</i></li> <li>• <i>Roles &amp; Role-Based Access Control</i></li> <li>• <i>Authorization Rules</i></li> <li>• <i>Views for Security</i></li> <li>• <i>SQL Authorization Commands</i></li> <li>• <i>Security Considerations</i></li> </ul> | <i>Exercises as given in the books</i>                   | <i>Connolly : Ch20</i>                             |
| 13 | 25<br>(Date) | <b>Indexing:</b> <ul style="list-style-type: none"> <li>• <i>Introduction to Indexing</i></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Exercises</i>                                         | <i>Connolly :</i>                                  |

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|    |              | <ul style="list-style-type: none"> <li>• <i>Purpose of Indexes</i></li> <li>• <i>Primary Indexes</i></li> <li>• <i>Secondary Indexes</i></li> <li>• <i>Dense &amp; Sparse Indexes</i></li> <li>• <i>Single-Level Indexes</i></li> <li>• <i>Multi-Level Indexes</i></li> <li>• <i>Index Selection &amp; Performance</i></li> <li>• <i>Advantages &amp; Disadvantages of Indexing</i></li> </ul>                                                                                                                                                                                                         | <i>as given in the books</i>                             | <i>Ch18</i>                   |
|    | 26<br>(Date) | <b>Introduction to NoSQL/Non-Relational Databases:</b> <ul style="list-style-type: none"> <li>• <i>Introduction to NoSQL Databases</i></li> <li>• <i>Need for NoSQL</i></li> <li>• <i>Characteristics of NoSQL Databases</i></li> <li>• <i>Relational vs. NoSQL Databases</i></li> <li>• <i>Types of NoSQL Databases</i></li> <li>• <i>NoSQL Data Models</i></li> <li>• <i>Advantages &amp; Limitations</i></li> <li>• <i>NoSQL Use Cases</i></li> <li>• <i>CAP Theorem Overview</i></li> </ul>                                                                                                        | <i>Exercises as given in the books</i>                   | <i>Shannon : Ch1</i>          |
| 14 | 27<br>(Date) | <b>MongoDB &amp; Document Model:</b> <ul style="list-style-type: none"> <li>• <i>Introduction to MongoDB</i></li> <li>• <i>MongoDB Architecture</i></li> <li>• <i>Document-Oriented Data Model</i></li> <li>• <i>Documents and Collections</i></li> <li>• <i>BSON Data Format</i></li> <li>• <i>MongoDB Data Types</i></li> <li>• <i>Embedded Documents &amp; Arrays</i></li> <li>• <i>MongoDB CRUD Operations</i></li> <li>• <i>MongoDB Querying</i></li> <li>• <i>MongoDB Use Cases</i></li> </ul>                                                                                                   | <i>Exercises as given in the books</i>                   | <i>Shannon : Ch2</i>          |
|    | 28<br>(Date) | <b>MongoDB Sharding &amp; Replication:</b> <ul style="list-style-type: none"> <li>• <i>MongoDB Replication</i></li> <li>• <i>Replica Sets</i></li> <li>• <i>Primary &amp; Secondary Nodes</i></li> <li>• <i>Automatic Failover</i></li> <li>• <i>Read &amp; Write Operations in Replication</i></li> <li>• <i>MongoDB Sharding</i></li> <li>• <i>Sharded Cluster Architecture</i></li> <li>• <i>Shard Keys</i></li> <li>• <i>Query Routing &amp; mongos</i></li> <li>• <i>Config Servers</i></li> <li>• <i>Balancing &amp; Data Distribution</i></li> <li>• <i>Sharding vs. Replication</i></li> </ul> | <i>Exercises as given in the books + Assignment No 4</i> | <i>Shannon : Ch9 and Ch13</i> |

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| 15 | 29<br>(Date) | <b>Database Transactions &amp; Properties:</b> <ul style="list-style-type: none"> <li>• Introduction to Database Transactions</li> <li>• Transaction Processing</li> <li>• Transaction States</li> <li>• ACID Properties</li> <li>• Atomicity</li> <li>• Consistency</li> <li>• Isolation</li> <li>• Durability</li> <li>• Importance of Transactions</li> <li>• Transaction Management in DBMS</li> </ul>                        | Exercises as given in the books | Connolly : Ch22 |
|    | 30<br>(Date) | <b>Transaction Management:</b> <ul style="list-style-type: none"> <li>• Concurrency Control</li> <li>• Need for Concurrency Control</li> <li>• Serial &amp; Concurrent Schedules</li> <li>• Serializability</li> <li>• Lock-Based Concurrency Control</li> <li>• Two-Phase Locking (2PL)</li> <li>• Deadlocks</li> <li>• Isolation Levels</li> <li>• Read Phenomena</li> <li>• Transaction Isolation &amp; Consistency</li> </ul> | Exercises as given in the books | Connolly : Ch22 |
| 16 | 31<br>(Date) | <b>Failure and Recovery in Transaction Management:</b> <ul style="list-style-type: none"> <li>• Types of Transaction Failures</li> <li>• System &amp; Media Failures</li> <li>• Recovery Concepts</li> <li>• Log-Based Recovery</li> <li>• Undo &amp; Redo</li> <li>• Checkpointing</li> <li>• Rollback &amp; Rollforward</li> <li>• Database Recovery Procedures</li> </ul>                                                      | Exercises as given in the books | Connolly : Ch22 |
|    | 32<br>(Date) | <ul style="list-style-type: none"> <li>• Review</li> </ul>                                                                                                                                                                                                                                                                                                                                                                        |                                 |                 |
| 17 | (Date)       |                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Final (50%)</b>              |                 |

**Note:** All Assignments/Quizes weightage is 25%